A POWER-SAVING ASYMPTOTIC FOR THE TWO-PARAMETER TYPE II SIEVE SUM OF THE ERDŐS–STRAUS EQUATION

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Abstract. We find the full asymptotic expansion, with a power-saving error term, of the weighted count of search pairs that controls a recent sieve bound for the Erdős–Straus equation. Its leading constant, three times the one previously proved, and its linear term agree with those found earlier by Liu; the constant term and the power-saving error are new.

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1. Introduction

The Erdős–Straus conjecture, posed in 1948, asserts that for every integer $n \geq 2$ the equation

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1.1}$$

has a solution in positive integers x, y, z [7, §D11]. A solution for n gives one for every multiple of n, so it is enough to treat primes, and polynomial identities settle every prime outside the six residue classes $1, 121, 169, 289, 361, 529$ modulo 840 [13]. These are the squares of the classes coprime to 840, and Mordell's barrier says that no identity of this kind can settle a square class, so the remaining primes cannot be reached by congruences alone.[1] The conjecture has been verified for all $n \leq 10^{17}$

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[1]More precisely, a polynomial identity solving (1.1) for all n in a residue class r modulo q exists only if r is not a square modulo q; see [6, Section 2] for a proof of this principle and for the families of identities behind it. The same obstruction is visible in practice: a search over the

by Salez [14] and for all $n \le 10^{18}$ by Mihnea and Dumitru [11], and Vaughan showed that the number of exceptions up to N is at most $N\exp(-c(\log N)^{2/3})$ [15]. Elsholtz and Tao [6] divided the solutions of (1.1) for a prime n into Type I, in which n divides exactly one of x, y, z, and Type II, in which it divides two of them, and determined the average number of solutions of each type. The conjecture remains open.

Dahan [5] recently asked how deep a search has to go before it finds a Type II solution. He showed that a prime n has a Type II solution if and only if there are positive integers u and a such that

$$an + u \text{ has a divisor } s \equiv -1 \pmod{4ua}, \tag{1.2}$$

and that one may take u and a coprime (Section 2). Let $E_2(N;J)$ be the set of primes $n \le N$ with $n \equiv 1 \pmod{24}$ for which (1.2) has no solution with coprime $u, a \le J$. Dahan bounds $|E_2(N;J)|$ by a sieve whose size is governed by the sum

$$W(J) = \sum_{\substack{u,a \le J \\ \gcd(u,a)=1}} \frac{1}{\varphi(4ua)}, \tag{1.3}$$

which counts the residue classes -1 modulo $4ua$ with the weight that each class receives among the primes.[2] Using $\varphi(m) \le m$ he proved $W(J) \ge (\kappa_1 + o(1))(\log J)^2$ with $\kappa_1 = 3/(2\pi^2) = 0.15198\ldots$ [5, Lemma 4.22]. He observed numerically that $W(J)/(\log J)^2$ seems to tend to a limit near 0.46, and the first of the questions with which he closes his paper is to prove $W(J) \ge (0.45 + o(1))(\log J)^2$ [5, Section 7]. We determine $W(J)$ up to an error that tends to zero.

Theorem 1.1. *Let*

$$\kappa = \frac{9}{2\pi^2} = 0.4559453263\ldots, \qquad T = \frac{\log 2}{3} + \sum_{p>2} \frac{\log p}{(p-1)(p^2+1)},$$

$$M = -\frac{4}{9}(\log 2)^2 - \sum_{p>2} (\log p)^2 \Big(\frac{p+1}{(p-1)(p^2+1)} + \frac{1}{(p-1)^2(p^2+1)^2} \Big),$$

the sums running over odd primes, and let γ be Euler's constant. Then, for $J \ge 2$,

$$W(J) = \kappa\big((\log J + \gamma + T)^2 + M\big) + O\big(J^{-1/2}\log J\big).$$

Numerically $T = 0.3131549610\ldots$ and $M = -1.0429405079\ldots$, so that

$$W(J) = \kappa(\log J)^2 + 0.8119206513\ldots \log J - 0.1140687011\ldots + O\big(J^{-1/2}\log J\big).$$

In particular $W(J) \sim \kappa(\log J)^2$ with $\kappa = 3\kappa_1$, and the limit near 0.46 that Dahan extrapolated from four values of $W(J)$ is $\kappa = 0.45594\ldots$. Liu [9] answered Dahan's question first: he proved $\sum_{u,a \le J,\,(u,a)=1} 1/\varphi(qua) = C(q)(\log J)^2 + B(q)\log J + O_q(1)$ for every fixed q, with $C(4) = \kappa$ and $B(4) = 2\kappa(\gamma + T)$. Theorem 1.1 adds the constant term $\kappa((\gamma+T)^2 + M)$ and replaces $O(1)$ by $O(J^{-1/2}\log J)$. Figure 1 compares the formula with $W(J)$ computed exactly.

Type II identities of Section 2 with small parameters leaves, among the integers below 10^8 in the six classes, exactly the perfect squares.

[2] By the prime number theorem in arithmetic progressions, the primes $p \le x$ with $p \equiv -1 \pmod{4ua}$ number about $\pi(x)/\varphi(4ua)$, so $W(J)$ is the expected number of pairs (u,a) for which a prime divisor of the right size lies in the right class. The use of the large sieve in this setting is in the spirit of [2], where it controls Liouville correlations uniformly in the coefficients.

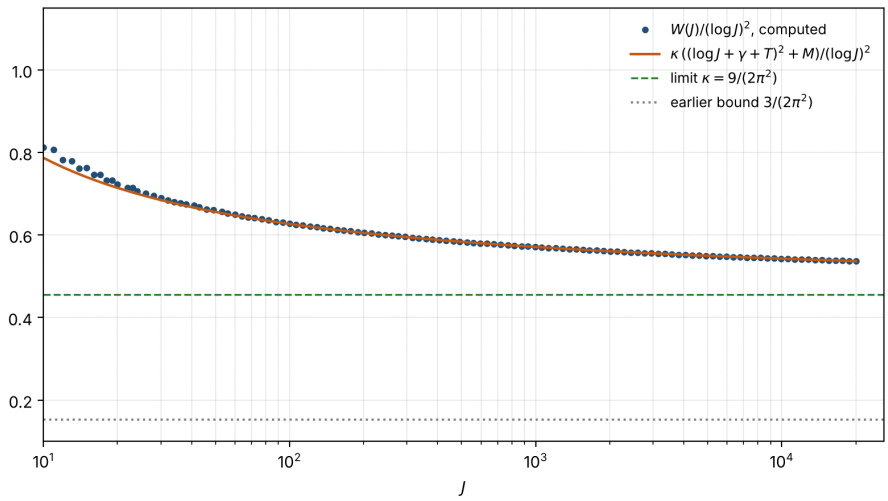


FIGURE 1. The ratio $W(J)/(\log J)^2$ computed from (1.3) for $10 \le J \le 2 \cdot 10^4$ (dots), the three-term formula of Theorem 1.1 divided by $(\log J)^2$ (solid line), its limit $\kappa = 9/(2\pi^2)$ (dashed) and the constant $3/(2\pi^2)$ proved in [5] (dotted). The ratio approaches κ slowly, as $1 + 2(\gamma + T)/\log J$, which is why it is still above 0.53 at $J = 2 \cdot 10^4$.

Dahan's sieve uses $W(J)$ only through a lower bound, and he notes that every constant of [5, Section 4.3] improves as soon as a sharper lower bound is proved [5, p. 28]. Substituting Theorem 1.1 into his arguments gives the following bounds; they rest on the arguments of [5], which we use as stated and have not verified beyond the step in which $W(J)$ enters.

Corollary 1.2. *Let $\kappa = 9/(2\pi^2)$. In the setting of* [5, Section 4.3]*:*

(1) *for fixed $\lambda \ge 1$ and $J = (\log N)^\lambda$,*

$$|E_2(N;J)| \ll N \exp\bigl(-(\kappa\lambda^2 + o(1))(\log\log N)^3\bigr),$$

against $\kappa_1\lambda^2$ in [5, Theorem 4.17]*;*

(2) *with $J = (\log N)^\lambda$ and $z = \exp(\sqrt{\log N})$ the exponent is*

$$(\kappa/2 + o(1))\lambda^2(\log\log N)^3,$$

against $\kappa_1/2$ in [5, Corollary 4.29]*;*

(3) *with the choices*

$$\log z = \Bigl(\frac{25e\log N}{\kappa}\Bigr)^{1/3} = 5.302\ldots(\log N)^{1/3},$$

$$\log J = \frac{\log z}{5\sqrt{e}} = 0.643\ldots(\log N)^{1/3},$$

$$|E_2(N;J)| \ll N \exp\bigl(-0.0943\,(\log N)^{2/3}\bigr),$$

against 0.065 *in* [5, Corollary 4.31].

The constant in (3) is $\frac{1}{2}(\kappa/(25e))^{1/3} = 0.09430385\ldots$, which is $3^{1/3}$ times Dahan's $0.06538\ldots$; Liu [9] obtained the same value from the leading constant. Corollary 1.2 bounds the number of primes for which no Type II solution is found among small parameters. A prime $n \equiv 1 \pmod{24}$ for which (1.1) has no solution lies in $E_2(N; J)$, so (3) also bounds the number of these primes, but it does not show that there are none. It does not prove the Erdős–Straus conjecture.

The proof of Theorem 1.1 rests on one observation: although $1/\varphi(4ua)$ is not a multiplicative function of the pair (u, a), the quantity $ua/\varphi(4ua)$ is, after the factor $\frac{1}{2}$ coming from the prime 2 is removed. Its Dirichlet inverse with respect to both variables is supported on squarefree pairs and decays fast enough to make a two-variable version of the familiar argument for $\sum_{n\le x} 1/\varphi(n)$ [12] work with a power saving. Section 2 recalls the criterion (1.2) with the identity behind it, Section 3 contains the multiplicative structure, Section 4 the proof, Section 5 the constants and the numerical comparison, and Section 7 the formal and numerical checks.

2. The Type II identity

The criterion (1.2) comes from an identity that we record because it gives the solution explicitly.

Lemma 2.1. *Let n, u, a be positive integers and let s be a divisor of $an + u$ with $s \equiv -1 \pmod{4ua}$. Write $s + 1 = 4ua\alpha$ and $an + u = sv$. Then*

$$\frac{4}{n} = \frac{1}{\alpha uv} + \frac{1}{a\alpha nu} + \frac{1}{a\alpha nv}. \tag{2.1}$$

Proof. The right-hand side equals $(an + u + v)/(a\alpha nuv)$. Since $an + u = sv = (4a\alpha u - 1)v$, the numerator is $4a\alpha uv$, and the fraction is $4/n$. □

For the prime $n = 409$ one may take $u = 1$, $a = 2$ and $s = 7$, which divides $819 = 7\cdot117$; then $\alpha = 1$, $v = 117$, and (2.1) reads $4/409 = 1/117+1/818+1/95706$. When n is prime and $n \nmid uv$, two of the three denominators in (2.1) are divisible by n and the third is not, so the solution is of Type II. Conversely every Type II solution arises in this way [5, Theorem 3.9], [6, Section 2]. If s is a witness for (u, a) and $d = \gcd(u, a)$, then s is coprime to d, because $d \mid s + 1$, and so s divides $(a/d)n + u/d$ and is $\equiv -1$ modulo $4(u/d)(a/d)$; hence the coprime pairs suffice [5, Lemma 3.10], which is why the sum (1.3) is restricted to them.[3]

3. A multiplicative function of two variables

For a positive integer m write m' for its largest odd divisor, and put $g(m) = m/\varphi(m)$.

Lemma 3.1. *For coprime positive integers u and a,*

$$\frac{ua}{\varphi(4ua)} = \frac{1}{2}\, g(u')\, g(a').$$

Proof. Write $u = 2^\beta u'$ and $a = 2^\delta a'$; since $\gcd(u, a) = 1$, the integers $2^{2+\beta+\delta}$, u', a' are pairwise coprime, so $\varphi(4ua) = 2^{1+\beta+\delta}\varphi(u')\varphi(a')$. Dividing $ua = 2^{\beta+\delta}u'a'$ by this gives the claim. □

[3] Figure 2 shows the terms of (1.3). A single prime $p \equiv -1 \pmod{4ua}$ dividing $an + u$ gives a witness, and the density of such primes is $1/\varphi(4ua)$; this is the source of the weight in (1.3).

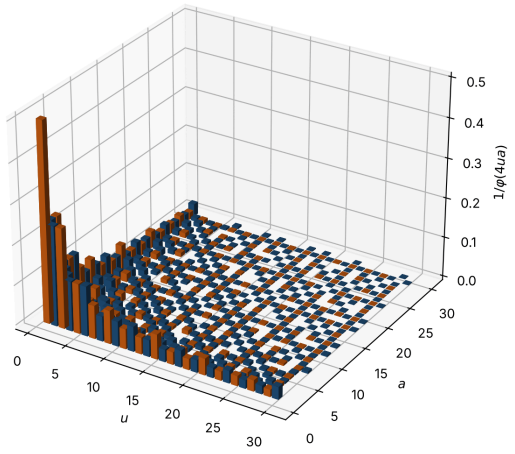


FIGURE 2. The terms $1/\varphi(4ua)$ of the sum (1.3) for coprime $u, a \le 30$; orange bars are the pairs with ua odd and blue bars those with ua even. Missing bars are the pairs with $\gcd(u,a) > 1$. By Lemma 3.1 an even product ua halves the weight, which is the source of the factor $3/4$ at the prime 2 in (3.2).

Consequently

$$W(J) = \frac{1}{2} \sum_{u,a \le J} \frac{h(u,a)}{ua}, \qquad h(u,a) = [\gcd(u,a) = 1]\, g(u')\, g(a'), \tag{3.1}$$

where $[\,\cdot\,]$ is 1 when the condition holds and 0 otherwise. The function h is multiplicative in the pair: $h(u_1u_2, a_1a_2) = h(u_1,a_1)h(u_2,a_2)$ whenever $\gcd(u_1a_1, u_2a_2) = 1$. It is therefore determined by its values $h(p^i,p^j)$, which are $h(1,1) = 1$; $h(p^i,p^j) = 0$ when $i,j \ge 1$; and, when exactly one of i,j is positive, $h(p^i,p^j) = p/(p-1)$ for odd p and $h(p^i,p^j) = 1$ for $p = 2$. Define k by Möbius inversion in both variables,

$$k(d,e) = \sum_{d_1d_2=d} \sum_{e_1e_2=e} \mu(d_2)\mu(e_2)\, h(d_1,e_1),$$

so that

$$h(u,a) = \sum_{d \mid u} \sum_{e \mid a} k(d,e).$$

The function k is again multiplicative in the pair, and $k(p^i,p^j) = h(p^i,p^j) - h(p^{i-1},p^j) - h(p^i,p^{j-1}) + h(p^{i-1},p^{j-1})$, with h set to 0 at negative exponents.

Lemma 3.2. *For an odd prime p,*

$$k(p,1) = k(1,p) = \frac{1}{p-1}, \qquad k(p,p) = -\frac{p+1}{p-1},$$

and $k(2,2) = -1$. At every prime, $k(p^i,p^j) = 0$ for all other pairs $(i,j) \ne (0,0)$.

Proof. For odd p, $k(p,1) = p/(p-1) - 1 = 1/(p-1)$ and $k(p,p) = 0 - 2p/(p-1) + 1 = -(p+1)/(p-1)$. If $i \ge 2$ and $j = 0$, the two values $h(p^i,1)$ and

$h(p^{i-1},1)$ are both $p/(p-1)$, and the other two vanish. If $i \geq 2$ and $j \geq 1$, then $h(p^i,p^j) = h(p^{i-1},p^j) = 0$ and $h(p^i,p^{j-1}) = h(p^{i-1},p^{j-1})$, both being 0 when $j \geq 2$ and $p/(p-1)$ when $j = 1$. The cases with i and j exchanged are symmetric. At $p = 2$ the same computation with $p/(p-1)$ replaced by 1 gives $k(2,1) = k(1,2) = 0$, $k(2,2) = 0 - 1 - 1 + 1 = -1$, and 0 elsewhere. □

Thus $k(d,e) \neq 0$ only when $d = q_1q_3$ and $e = q_2q_3$ with q_1, q_2, q_3 squarefree and pairwise coprime and q_1q_2 odd. For real σ, τ consider

$$Z(\sigma,\tau) = \sum_{d,e\geq 1} \frac{k(d,e)}{d^{1+\sigma}e^{1+\tau}} = \prod_p E_p(\sigma,\tau),$$

where, for odd p,

$$E_p(\sigma,\tau) = 1 + \frac{1}{(p-1)p^{1+\sigma}} + \frac{1}{(p-1)p^{1+\tau}} - \frac{p+1}{(p-1)p^{2+\sigma+\tau}},$$

and $E_2(\sigma,\tau) = 1 - 2^{-2-\sigma-\tau}$. The series converges absolutely when $\sigma > -1$, $\tau > -1$ and $\sigma + \tau > -1$, because then the same product with every term replaced by its absolute value converges: its local factor is $1 + O(p^{-2-\sigma} + p^{-2-\tau} + p^{-2-\sigma-\tau})$ and each exponent exceeds 1. In this open region Z is a convergent Euler product of nonvanishing factors, and the derivatives below may be taken term by term. We use the region at the points $(0,0)$ and $(-\frac{1}{2},0)$, $(0,-\frac{1}{2})$; in particular the constant

$$A = \sum_{d,e} \frac{|k(d,e)|\,(1+\log d)^2(1+\log e)^2}{d^{1/2}e}$$

and its mirror image are finite.

Lemma 3.3.

$$Z(0,0) = \frac{3}{4}\prod_{p>2}\Big(1 + \frac{1}{p^2}\Big) = \frac{9}{\pi^2}, \tag{3.2}$$

$$\frac{\partial}{\partial\sigma}\log Z(0,0) = \frac{\partial}{\partial\tau}\log Z(0,0) = T, \tag{3.3}$$

$$\frac{\partial^2}{\partial\sigma\,\partial\tau}\log Z(0,0) = M. \tag{3.4}$$

Proof. For odd p,

$$E_p(0,0) = 1 + \frac{2}{p(p-1)} - \frac{p+1}{(p-1)p^2} = 1 + \frac{2p-p-1}{(p-1)p^2} = 1 + \frac{1}{p^2},$$

and $E_2(0,0) = 3/4$. Since $\prod_p(1+p^{-2}) = \zeta(2)/\zeta(4) = 15/\pi^2$ and the factor at $p = 2$ is $5/4$, we obtain $Z(0,0) = \frac{3}{4}\cdot\frac{4}{5}\cdot\frac{15}{\pi^2} = \frac{9}{\pi^2}$. For the derivatives, write $\ell = \log p$. At odd p,

$$\partial_\sigma E_p(0,0) = -\frac{\ell}{(p-1)p} + \frac{(p+1)\ell}{(p-1)p^2} = \frac{\ell}{(p-1)p^2}, \qquad \partial_\sigma\partial_\tau E_p(0,0) = -\frac{(p+1)\ell^2}{(p-1)p^2},$$

and dividing by $E_p(0,0) = (p^2+1)/p^2$ gives $\partial_\sigma \log E_p = \ell/((p-1)(p^2+1))$ and

$$\partial_\sigma\partial_\tau \log E_p = \frac{\partial_\sigma\partial_\tau E_p}{E_p} - \frac{\partial_\sigma E_p\,\partial_\tau E_p}{E_p^2} = -\ell^2\Big(\frac{p+1}{(p-1)(p^2+1)} + \frac{1}{(p-1)^2(p^2+1)^2}\Big).$$

At $p = 2$, with $w = 2^{-2-\sigma-\tau}$ we have $\log E_2 = \log(1-w)$, $\partial_\sigma \log E_2 = w\log 2/(1-w)$ and $\partial_\sigma\partial_\tau \log E_2 = -w(\log 2)^2/(1-w)^2$; at $w = 1/4$ these are $(\log 2)/3$ and

$-\frac{4}{9}(\log 2)^2$. Summing over p gives (3.3) and (3.4); the symmetry $E_p(\sigma,\tau) = E_p(\tau,\sigma)$ gives the equality of the two first derivatives. □

Figure 3 shows the surface $\log Z$ near the point $(0,0)$; the theorem reads off its height, its two slopes and its mixed curvature there.

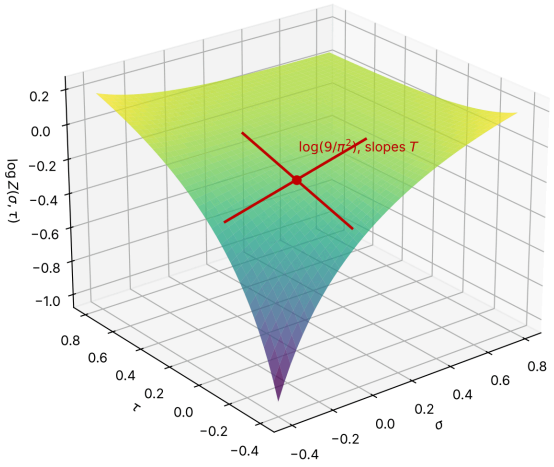


FIGURE 3. The surface $(\sigma,\tau) \mapsto \log Z(\sigma,\tau)$ over $-0.4 \le \sigma,\tau \le 0.8$, computed from the Euler product over the primes below $2\cdot 10^5$. The marked point is $(0, 0, \log(9/\pi^2))$, which gives the leading constant $\kappa = Z(0,0)/2$; the two lines through it have slope T in the σ and τ directions, and the mixed second derivative there is M. The surface drops steeply towards the corner $\sigma = \tau = -\frac{1}{2}$, where the line $\sigma + \tau = -1$ bounds the region of absolute convergence.

4. Proof of Theorem 1.1

Let $H(x) = \sum_{m \le x} 1/m$ for $x \ge 1$. By (3.1) and the inversion formula,

$$W(J) = \frac{1}{2} \sum_{u,a \le J} \frac{1}{ua} \sum_{d \mid u,\ e \mid a} k(d,e) = \frac{1}{2} \sum_{d,e \le J} \frac{k(d,e)}{de}\, H\Big(\frac{J}{d}\Big) H\Big(\frac{J}{e}\Big),$$

on writing $u = dm$, $a = en$. We use $H(x) = \log x + \gamma + R(x)$ with $|R(x)| \le 2/x$ for $x \ge 1$ (Appendix A.4).[4] Put $L = \log J + \gamma$. Then $H(J/d) = L - \log d + R(J/d)$, and

$$H\Big(\frac{J}{d}\Big) H\Big(\frac{J}{e}\Big) = (L - \log d)(L - \log e) + R\Big(\frac{J}{d}\Big) H\Big(\frac{J}{e}\Big) + R\Big(\frac{J}{e}\Big)(L - \log d).$$

[4]The constant γ enters only through this expansion, and it is the reason why Euler's constant appears beside the arithmetic constant T in Theorem 1.1: each of the two harmonic sums contributes one γ, and the cross term $2\gamma T$ in $(\log J + \gamma + T)^2$ couples the archimedean part of the problem with the primes.

The remainder terms. For $d, e \le J$ we have $0 \le H(J/e) \le 1+\log J$, $|L-\log d| \le 1+\log J$ and $|R(J/d)| \le 2d/J \le 2(d/J)^{1/2}$. Hence the two remainder terms, after multiplication by $\frac{1}{2}|k(d,e)|/(de)$ and summation, contribute at most

$$\frac{1+\log J}{J^{1/2}} \sum_{d,e} \frac{|k(d,e)|}{d^{1/2}e} + \frac{1+\log J}{J^{1/2}} \sum_{d,e} \frac{|k(d,e)|}{de^{1/2}} \le \frac{2A(1+\log J)}{J^{1/2}}.$$

Completing the main term. It remains to replace $\sum_{d,e\le J}$ by $\sum_{d,e\ge 1}$ in

$$\frac{1}{2} \sum_{d,e} \frac{k(d,e)}{de} (L-\log d)(L-\log e).$$

The pairs with $d > J$ satisfy $\log J < \log d$, so $|L-\log d| \le 1+2\log d$ and $|L-\log e| \le 1+\log d+\log e$, and $1/d \le J^{-1/2} d^{-1/2}$; their contribution is therefore at most

$$\frac{1}{J^{1/2}} \sum_{d,e} \frac{|k(d,e)|(1+2\log d)(1+\log d+\log e)}{d^{1/2}e} \le \frac{2A}{J^{1/2}}.$$

The pairs with $e > J$ are treated in the same way, and the pairs with both $d, e > J$ are counted twice, which only helps the bound. So

$$W(J) = \frac{1}{2}\big(S_0 L^2 - 2S_1 L + S_{11}\big) + O\big(J^{-1/2}\log J\big),$$

with

$$S_0 = \sum_{d,e} \frac{k(d,e)}{de}, \qquad S_1 = \sum_{d,e} \frac{k(d,e)\log d}{de} = \sum_{d,e} \frac{k(d,e)\log e}{de},$$

$$S_{11} = \sum_{d,e} \frac{k(d,e)\log d \log e}{de}.$$

In terms of Z these are $S_0 = Z$, $S_1 = -\partial_\sigma Z$ and $S_{11} = \partial_\sigma \partial_\tau Z$, all at $(0,0)$. By Lemma 3.3, $\partial_\sigma Z / Z = T$ and

$$\frac{\partial_\sigma \partial_\tau Z}{Z} = \partial_\sigma \partial_\tau \log Z + \frac{\partial_\sigma Z}{Z} \cdot \frac{\partial_\tau Z}{Z} = M + T^2.$$

Therefore

$$\frac{1}{2}\big(S_0 L^2 - 2S_1 L + S_{11}\big) = \frac{Z(0,0)}{2}\big(L^2 + 2TL + T^2 + M\big) = \frac{9}{2\pi^2}\big((L+T)^2 + M\big),$$

which is Theorem 1.1. Expanding the square gives the second form, with the coefficient of $\log J$ equal to $2\kappa(\gamma+T)$ and the constant equal to $\kappa((\gamma+T)^2+M)$. □

Remark 4.1. The same argument applies to the one-parameter sum $\sum_{u\le J} 1/\varphi(4u)$ of [5, Lemma 4.21], where $4u/\varphi(4u) = 2g(u')$ and one recovers the constant $\kappa_0 = \zeta(2)\zeta(3)/(3\zeta(6))$ of that lemma. The two-variable case differs only in the diagonal value $k(p,p)$, which comes from the coprimality condition and produces the factor $1+p^{-2}$ in place of $1+1/(p(p-1))$.[5]

[5]The Euler factors $1+p^{-2}$ and the quotient $\zeta(2)/\zeta(4)$ that they produce are typical of sums over coprime pairs; a different instance, where the same quotient appears through Ramanujan's function, is studied in [3]. The two-variable inversion used here is the Dirichlet-series counterpart of the approximation arguments of [4], where a single-variable Möbius sum has to be controlled with an explicit error.

5. The constants

The series for T converges like $\sum_p (\log p) p^{-3}$, and summing over the primes below 10^7 determines it to twelve digits. The series for M converges more slowly, like $\sum_p (\log p)^2 p^{-2}$: by the prime number theorem the primes above P contribute $(\log P + 1)/P$ up to a factor $1 + o(1)$, which is $1.7 \cdot 10^{-6}$ at $P = 10^7$.[6] Summing up to $4 \cdot 10^8$ and adding this correction gives

$$T = 0.3131549610\ldots, \qquad M = -1.0429405079\ldots, \qquad \gamma + T = 0.8903706259\ldots,$$

and hence $2\kappa(\gamma + T) = 0.8119206513\ldots$ and $\kappa((\gamma + T)^2 + M) = -0.1140687011\ldots$. Table 1 compares the formula with $W(J)$ computed directly from (1.3). The first four values of the ratio $W(J)/(\log J)^2$ are those reported in [5, Section 6, item (8)].

Table 1. The sum $W(J)$, the formula $\kappa((\log J + \gamma + T)^2 + M)$ of Theorem 1.1, their difference, and the ratio $W(J)/(\log J)^2$.

J	$W(J)$	formula	difference	$W(J)/(\log J)^2$
100	13.317254584	13.294466733	$2.28 \cdot 10^{-2}$	0.62795
300	19.358190414	19.350280717	$7.91 \cdot 10^{-3}$	0.59503
1000	27.253212761	27.250861444	$2.35 \cdot 10^{-3}$	0.57114
3000	35.614356280	35.613438048	$9.18 \cdot 10^{-4}$	0.55559
10000	46.042326377	46.042007484	$3.19 \cdot 10^{-4}$	0.54276
20000	52.645668414	52.645468987	$1.99 \cdot 10^{-4}$	0.53677

The difference decays roughly like J^{-1}, faster than the bound $J^{-1/2} \log J$ of Theorem 1.1.[7] The four values of the ratio $W(J)/(\log J)^2$ at $J = 100, 300, 1000, 3000$ were fitted in [5] against $\kappa^* + B_0/\log J$, giving $\kappa^* \approx 0.4576$ and $B_0 \approx 0.784$. Theorem 1.1 explains both numbers: the ratio equals $\kappa + 2\kappa(\gamma + T)/\log J + O((\log J)^{-2})$, and $2\kappa(\gamma + T) = 0.812$, so a two-term fit over this range absorbs part of the $(\log J)^{-2}$ term into its constant and lands slightly above $\kappa = 0.45594$.

6. Proof of Corollary 1.2

In [5] the sum $W(J)$ enters only through [5, Lemma 4.22], the lower bound $W(J) \geq (\kappa_1 + o(1))(\log J)^2$, which is used in [5, Lemma 4.23] to bound from below the sieve quantity $S(J, z)$ and through it in Lemmas 4.24 and 4.30, Theorem 4.17 and Corollaries 4.29 and 4.31 of [5]. Theorem 1.1 gives $W(J) \geq (\kappa + o(1))(\log J)^2$, and none of these arguments uses any other property of κ_1, so each of them holds with κ_1 replaced by κ. This gives (1) and (2) directly. For (3), the proof of [5, Corollary 4.31] shows that the saturation point $L_\star = \log z$ satisfies $\kappa L_\star^3/(50e) \leq \frac{1}{2} \log N + O(1)$, so that $L_\star \leq (1 + o(1))(25e \log N/\kappa)^{1/3}$, and that the sieve quantity

[6] Summing over the primes up to $P = 10^7$, 10^8 and $4 \cdot 10^8$ gives -1.0429387961, -1.0429403137 and -1.0429404559; subtracting $(\log P + 1)/P$ from each gives -1.04294050791, -1.04294050795 and -1.04294050794. The agreement of the corrected values to eleven digits, while the uncorrected ones move in the sixth, is the evidence for the digits of M stated in the text.

[7] The exponent $\frac{1}{2}$ comes from Rankin's trick with the exponent $\frac{1}{2}$, which is the most that the condition $\sigma > -1$ allows on the line $\tau = 0$ after the pairs with $d > J$ are bounded crudely. Any exponent below 1 can be reached in the same way, at the cost of a larger constant; we do not pursue this, since only the main term enters the applications.

is $S = (\frac{1}{2} + o(1)) \log N/L_\star \geq (\frac{1}{2} + o(1))(\kappa/(25e))^{1/3}(\log N)^{2/3}$; the exceptional set is $\ll N \exp(-(1 + o(1))S)$ [5, Theorem 4.28]. With $\kappa = 9/(2\pi^2)$ the constant is $0.0943\ldots$, and $\log J = L_\star/(5\sqrt{e}) \leq 0.644(\log N)^{1/3}$, so the condition $J^5 < z$ and the bound $J \leq \exp((\log N)^{1/2})$ required there continue to hold. □

7. Formal and numerical verification

The algebraic steps of the proof are checked in Lean 4 [8] with Mathlib [10]: the factorisation $\varphi(4ua) = c\,\varphi(u)\varphi(a)$ with $c = 4$ when ua is even and $c = 2$ when ua is odd, from which Lemma 3.1 follows; the local values of Lemma 3.2; the local factors and derivatives of Lemma 3.3 as identities in the variable p; the value $Z(0, 0) = 9/\pi^2$ from $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$; the identity (2.1); and the example $n = 409$. All these theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The analytic estimates of Section 4 are not formalised. Independent programs in C, C++, Python and Julia recompute $W(J)$ for $J \leq 2 \cdot 10^4$ and compare it with the formula, check in exact rational arithmetic that $h(u, a) = \sum_{d|u,\, e|a} k(d, e)$ for all $u, a \leq 400$, recompute the local factors exactly for every prime below 10^4, and evaluate T, M and the constants of Corollary 1.2.

Appendix A. Detailed derivations

This appendix writes out every computation used in Sections 3 to 4 and in Corollary 1.2, so that the paper can be checked line by line without outside sources beyond the prime number theorem in its weakest form and the values $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$.

A.1. **The totient of $4n$.** Write $n = 2^\beta n'$ with n' odd. Since φ is multiplicative and $\varphi(2^k) = 2^{k-1}$ for $k \geq 1$,

$$\varphi(4n) = \varphi(2^{2+\beta})\varphi(n') = 2^{1+\beta}\varphi(n').$$

If n is odd, $\beta = 0$ and $\varphi(4n) = 2\varphi(n)$. If n is even, $\varphi(n) = 2^{\beta-1}\varphi(n')$, so $\varphi(4n) = 4\varphi(n)$. For coprime u and a we have $\varphi(ua) = \varphi(u)\varphi(a)$, hence

$$\varphi(4ua) = c\,\varphi(u)\varphi(a),$$

with $c = 2$ when ua is odd and $c = 4$ when ua is even. This is the form checked in Lean. Lemma 3.1 is the same statement written with u' and a': if $u = 2^\beta u'$ then $u/\varphi(u) = 2\,g(u')$ when $\beta \geq 1$ and $g(u')$ when $\beta = 0$, and the factors of 2 combine with c to give $\frac{1}{2}$ in both cases.[8]

A.2. **The local table of h and k.** At an odd prime p put $r = p/(p - 1)$. The values $h(p^i, p^j)$ for $0 \leq i, j \leq 2$ are

	$j = 0$	$j = 1$	$j = 2$
$i = 0$	1	r	r
$i = 1$	r	0	0
$i = 2$	r	0	0

and the values of $k(p^i, p^j) = h(p^i, p^j) - h(p^{i-1}, p^j) - h(p^i, p^{j-1}) + h(p^{i-1}, p^{j-1})$ are

[8] Explicitly, when ua is odd, $ua/\varphi(4ua) = ua/(2\varphi(u)\varphi(a)) = \frac{1}{2}g(u)g(a)$; when, say, u is even and a odd, $ua/\varphi(4ua) = ua/(4\varphi(u)\varphi(a)) = \frac{1}{4} \cdot 2g(u') \cdot g(a) = \frac{1}{2}g(u')g(a')$.

	$j=0$	$j=1$	$j=2$
$i=0$	1	$r-1$	0
$i=1$	$r-1$	$1-2r$	0
$i=2$	0	0	0

with $r-1=1/(p-1)$ and $1-2r=-(p+1)/(p-1)$. The table continues with zeros because every row and every column of h is constant from the second entry on. At $p=2$ one replaces r by 1, which gives $k(2,1)=k(1,2)=0$ and $k(2,2)=-1$. As a check, $\sum_{i,j}k(p^i,p^j)=h(p^\infty,p^\infty)$ in the obvious sense: $1+2(r-1)+(1-2r)=0$, which is the value of h on pairs with both exponents positive.

A.3. **Absolute convergence.** For real σ,τ the majorant of Z is $\prod_p \bar{E}_p$ with

$$\bar{E}_p = 1+\frac{1}{(p-1)p^{1+\sigma}}+\frac{1}{(p-1)p^{1+\tau}}+\frac{p+1}{(p-1)p^{2+\sigma+\tau}} \qquad (p>2)$$

and $\bar{E}_2=1+2^{-2-\sigma-\tau}$. Since $1/(p-1)\le 2/p$ and $(p+1)/(p-1)\le 2$, we have $\bar{E}_p-1\le 2\big(p^{-2-\sigma}+p^{-2-\tau}+p^{-2-\sigma-\tau}\big)$, and the product converges when the three exponents $2+\sigma$, $2+\tau$ and $2+\sigma+\tau$ all exceed 1. A product of factors $\bar{E}_p\ge 1$ converges if and only if $\sum(\bar{E}_p-1)$ does, and then $\sum_{d,e}|k(d,e)|d^{-1-\sigma}e^{-1-\tau}=\prod_p\bar{E}_p$ by multiplicativity. Multiplying by $(1+\log d)^2(1+\log e)^2$ does not affect convergence at an interior point, because $(1+\log d)^2\le C_\varepsilon d^\varepsilon$ for each $\varepsilon>0$; at $(\sigma,\tau)=(-\frac12,0)$ one may take $\varepsilon=\frac18$, which moves the point to $(-\frac38,\frac18)$, still inside the region. This proves that the constant A of Section 3 is finite.[9]

A.4. **The harmonic sum.** For integers $m\ge 1$, $H(m)-\log m-\gamma=\frac{1}{2m}-\frac{1}{12m^2}+\theta\frac{1}{120m^4}$ with $0<\theta<1$, which lies in $(0,\frac{1}{2m})$. For $m\le x<m+1$, $H(x)=H(m)$ and $\log x-\log m\in[0,\log(1+\frac1m))\subset[0,\frac1m)$. Hence $R(x)=H(x)-\log x-\gamma$ lies in $(-\frac1m,\frac{1}{2m})$, so $|R(x)|<1/m$; since $m>x-1\ge x/2$ for $x\ge 2$, and $m=1$ for $1\le x<2$, we get $|R(x)|\le 2/x$. This is the bound used in Section 4.

A.5. **The derivatives at** $(0,0)$**.** Write $x=p^{-\sigma}$ and $y=p^{-\tau}$, so that, at an odd prime,

$$E_p=1+\frac{x}{(p-1)p}+\frac{y}{(p-1)p}-\frac{(p+1)xy}{(p-1)p^2},$$

and $\partial_\sigma=-\ell x\,\partial_x$, $\partial_\tau=-\ell y\,\partial_y$ with $\ell=\log p$. At $x=y=1$:

$$\partial_\sigma E_p=-\ell\Big(\frac{1}{(p-1)p}-\frac{p+1}{(p-1)p^2}\Big)=-\ell\,\frac{p-(p+1)}{(p-1)p^2}=\frac{\ell}{(p-1)p^2},$$

$$\partial_\sigma\partial_\tau E_p=\ell^2xy\,\partial_x\partial_y E_p=-\frac{(p+1)\ell^2}{(p-1)p^2}.$$

With $E_p=(p^2+1)/p^2$,

$$\frac{\partial_\sigma E_p}{E_p}=\frac{\ell}{(p-1)(p^2+1)},\qquad \frac{\partial_\sigma\partial_\tau E_p}{E_p}=-\frac{(p+1)\ell^2}{(p-1)(p^2+1)},$$

$$\frac{\partial_\sigma E_p\,\partial_\tau E_p}{E_p^2}=\frac{\ell^2}{(p-1)^2(p^2+1)^2}.$$

[9]Numerically, $\sum_{d,e}|k(d,e)|d^{-1/2}e^{-1}=\prod_p\bar{E}_p(-\frac12,0)$ is about 5.4; the logarithmic weights increase the constant but play no role in the exponent of J.

The terms are $O(\ell p^{-3})$ and $O(\ell^2 p^{-2})$, and they remain so uniformly in a neighbourhood of $(0,0)$, so the series of derivatives converge uniformly there and $\log Z = \sum_p \log E_p$ may be differentiated term by term. These are the identities `local_T` and `local_M` of the Lean file, stated for a real variable $p > 1$.

A.6. **From Z to the three-term formula.** Differentiating the series $Z(\sigma,\tau) = \sum k(d,e) d^{-1-\sigma} e^{-1-\tau}$ term by term gives

$$\partial_\sigma Z(0,0) = -\sum \frac{k(d,e)\log d}{de} = -S_1,$$

$$\partial_\sigma \partial_\tau Z(0,0) = \sum \frac{k(d,e)\log d \log e}{de} = S_{11}.$$

Then

$$S_0 L^2 - 2S_1 L + S_{11} = Z\Big(L^2 + 2L\frac{\partial_\sigma Z}{Z} + \frac{\partial_\sigma\partial_\tau Z}{Z}\Big) = Z\big(L^2 + 2LT + T^2 + M\big),$$

which is $Z\big((L+T)^2 + M\big)$, using $\partial_\sigma\partial_\tau \log Z = \partial_\sigma\partial_\tau Z/Z - (\partial_\sigma Z/Z)(\partial_\tau Z/Z)$. With $L = \log J + \gamma$,

$$\frac{Z}{2}\big((L+T)^2 + M\big) = \kappa(\log J)^2 + 2\kappa(\gamma+T)\log J + \kappa\big((\gamma+T)^2 + M\big).$$

Numerically $\gamma + T = 0.5772156649 + 0.3131549610 = 0.8903706259$, $2\kappa(\gamma+T) = 0.8119206513$, $(\gamma+T)^2 = 0.7927598515$ and $\kappa(0.7927598515 - 1.0429405079) = -0.1140687011$.

A.7. **The constants of Corollary 1.2.** With $\kappa = 0.4559453264$ and $25e = 67.957045711$,

$$\Big(\frac{25e}{\kappa}\Big)^{1/3} = (149.0464\ldots)^{1/3} = 5.3020104,$$

$$\frac{5.3020104}{5\sqrt{e}} = \frac{5.3020104}{8.2436064} = 0.6431664,$$

$$\frac{1}{2}\Big(\frac{\kappa}{25e}\Big)^{1/3} = \frac{1}{2}\cdot 0.1886077 = 0.0943039.$$

For $\kappa_1 = 3/(2\pi^2) = 0.1519818$ the same expressions give 7.6468222 and 0.0653866, the values of [5, Corollary 4.31]. The ratio of the two exponents is $(\kappa/\kappa_1)^{1/3} = 3^{1/3} = 1.4422496$, and the ratio of the constants in parts (1) and (2) is $\kappa/\kappa_1 = 3$.

Data and code availability. The LaTeX source of the paper, the Lean 4 formalisation described in Section 7, and the C, C++, Python, Julia and shell programs that recompute the quantities of Sections 5 and 7 are available at `https://github.com/creelie/ErdosProblem1` [1]; the paper is available at `https://github.com/creelie/ErdosProblem1/blob/main/preprintErdosStrausSieveConstant/ErdosStrausSieveConstant.pdf`. Deep Bhattacharjee carried out the computations and the Lean, Julia, C, C++, Python and shell verifications, and the proofs and the text were drafted, with AI assistance (Claude, Anthropic); both authors approved the final version.

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